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VNUHCM - UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

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PROJECT IDEA &

TEAM CONTRACT

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Class: 24C05

Team: 08

24C05

Ho Chi Minh City

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1. Team Information

1.1 Team

Item	Value
Team ID	T08
Team name	Ne1ntendo
Team lead	Hoàng Minh Huy
Communication channel	https://discord.gg/swvV6C7j9

1.2 Members

Student ID	Full name	Email	Main role	Backup role
24127278	Hoàng Minh Huy	hmhuy2428@clc.fitus.edu.vn	<Team lead / PM>	<Developer>
24127048	Nguyễn Thế Huy	nthuy2406@clc.fitus.edu.vn	<Business analyst>	<Designer>
24127450	Nguyễn Chí Minh	ncminh2419@clc.fitus.edu.vn	<Designer>	<Tester>
24127329	Lê Nguyễn Quốc Bảo	lnqbao2418@clc.fitus.edu.vn	<Developer>	<Team lead / PM>
24127142	Đoàn Minh Tuấn	dmtuan2415@clc.fitus.edu.vn	<Tester>	<Business analyst>

2. Project Idea Overview

2.1 Project Name and One-Sentence Pitch

InfraLine - A web app project that allows both students/employees to report infrastructure problems such as faulty lights, air conditioners, projectors, etc., to the facility's technical support to help organizations oversee, evaluate, and manage their infrastructure efficiently.

2.2 Problem and Motivation

Infrastructures like machines in companies and appliances in schools are often parts of a larger process (e.g., a conveyor belt transports products to a robot arm or a projector displays theory videos and footage for teaching purposes). If one or many of the infrastructure components break down, it will be a hassle for both the user (employees and students) to manually write emails or messages locating/describing the problem and the technical manager who has to manually manage many flows of reports and emails to evaluate which one should be fixed first, where to fix it, as well as the urgency of the problem.

That's where InfraLine comes in. InfraLine will act like a management system for infrastructure problems in your facility, allowing users to easily report the status of the infrastructure as well as the problem it's currently having while merging multiple flows of issue reports into a single flow, helping the technical support team to easily locate and handle it. Furthermore, administrators can see the statistics and categories as well as set priorities for critical infrastructure.

2.3 Target Users

User group	What they need from the system
Student/Employees	Report a broken light, a faulty projector in a classroom, damaged machines in a factory, an office, or other areas. See the status of each report and whether the problem has been fixed.

Maintenance	See the reports the system assigns to them, update the progress, and write a note once the repair is done.
Administrator	Manage locations, categories, and accounts. View statistics by status, category, and location. See which reports are still open and set priorities for upcoming tasks.

2.4 Main Features

#	Feature	Short description	Must-have?
1	Broken infrastructure report	Users can upload a post to report broken infrastructure on a forum for the maintenance team to see.	Yes
2	Communications channel	A maintenance team member can enter a private chatbox with the problem's poster to get a better understanding of the problem through back-and-forth chatting.	No
3	Location ping location	Users can narrow down the location on a dropdown box where the broken infrastructure is located with a picture or short description of the fault.	Yes
4	Maintenance history	Users and administrators can see fixed infrastructure, who fixed it, and when the problem was dealt with.	No
5	Maintenance status	Users and administrators can see the current status of the problem, whether it's addressed, still being worked on, or no response.	Yes
6	Maintenance team checklist	Maintenance team members will have a checklist of broken infrastructure they are tasked to deal with.	Yes
7	Email response	An automatic email response will be sent to the post's author whenever the problem has been dealt with.	No
8	Broken infrastructure post tags	Users and administrators can apply tags to their posts to specify whether the broken infrastructure is mechanical, electrical, critical,... so the maintenance team has an easier time judging and prioritizing tasks.	Yes
9	Set task priority	Administrators can set priority to problems posted on the forum.	Yes
10	Problem moderation	Administrators can approve or reject a post if they feel the problem isn't worth addressing or is outside the scope of the maintenance team.	Yes

2.5 Similar Products

MaintainX: A CMMS that covers a wide range of maintenance and asset management issues, including preventative maintenance and inventory tracking, and supports multiple enterprise sites. InfraLine focuses on issue reporting and work order management in a single facility.

FixMyStreet: An open-source framework that allows anyone to run a website for reporting street problems in their country and directing that information to relevant authorities. InfraLine is a standalone software that focuses on workplace infrastructure and includes work order management alongside reporting.

Snap Send Solve: An app for reporting issues in Australian and New Zealand communities such as degraded footpaths and fallen branches, where designated solvers can go out and fix those issues. InfraLine focuses on workplace issues and can be used anywhere in the world as it is confined to single facilities.

2.6 Platform and Technology Ideas

- Our intention for InfraLine is a web application:
    - **Languages:** Python, HTML/CSS, JavaScript, PostgresSQL.
    - **Framework:**
        - Back-end: Django (Python)
        - Front-end: React.js
    - **Database:** Postgres

2.7 Feasibility, Scope and Open Questions

The must-have features are divided into 3 groups: user, maintenance team, admin communicating through problem posts. If working simultaneously on these three groups, a team of 4-5 members may complete the project within 3 weeks with proper planning.

First of all, we just focus on responsiveness on the web to save our time, so mobile apps are out of scope in this project. Moreover, for an app like this, an AI assistant in assessing and specifying the problem would be useful, but due to the restriction of time and resources, this feature is sadly out of scope for our team. Finally,

we decided not to make a 2D/3D floor map to show where the infrastructure is broken. Instead, the user selects their location through 3 dependent dropdown menus (campus -> block -> floor -> room). Selecting one level filters the options of the next level.

Our lack of knowledge in cybersecurity might expose some weaknesses to be exploited; synchronization in real time of multiple users must also be considered, yet our team has little experience in this field, which might cause negative feedback for user experience. In some cases, we also worry about the images uploaded to the web, which may put pressure on server memory.

The questions we want to ask TA: Can this project be considered a complete system, and is it qualified to be included in a CV?; Are we required to integrate institutional SSO (Google/Microsoft OAuth via email), or is standard Email/Password authentication?

3. Team Contract

3.1 Goals and Expectations

This project is important for our major and career path, so we intend to go in with a serious attempt. For that reason, our team is aiming for 90-100% of project grade.

And to do that, we plan to spend about 2 hours on 3 weekdays and 2 hours on weekends, which makes it approximately 10 hours per week. This number is made by considering both the time for this subject as well as other subjects in this semester, and it may also vary from week to week due to multiple deadlines.

3.2 Communication

- ✔ Main channel for discussion: https://discord.gg/swvV6C7j9
- ✔ Expected response time to messages: Within 12 hours on weekdays, within 6 hours on weekends.
- ✔ Where decisions and files are kept: GitHub repository, Discord channel.

3.3 Meetings

- ✔ Weekly status meeting with the TA: Friday, 3 PM, Online
- ✔ Internal team meetings: Any day of the week depending on the plan, 8 PM.
- ✔ Attendance rule: Inform the team a day in advance if you cannot attend; reason of absence will also be taken into consideration; being late by 30 minutes or more is considered absent; missed meetings are recorded in the weekly status report.
- ✔ Weekly status report deadline chosen by the team: Tuesday, 10 PM.

3.4 Roles and Responsibilities

- **For extra responsibilities:**

Student ID	Full name	Main role	Backup role	Extra responsibilities
24127278	Hoàng Minh Huy	<Team lead / PM>	<Developer>	Handle Git merging and conflicts + improvise for all other team members.
24127048	Nguyễn Thế Huy	<Business analyst>	<Designer>	Prepare presentation + Code documentation
24127450	Nguyễn Chí Minh	<Designer>	<Tester>	Data + User experience feedback.
24127329	Lê Nguyễn Quốc Bảo	<Developer>	<Team lead / PM>	Set up structural code templates + Technical advisors
24127142	Đoàn Minh Tuấn	<Tester>	<Business analyst>	Set up CI/CD for the project.

3.5 Work Standards

✔ **Internal deadline:** For our team, we plan to always finish the PA one day before the deadline, and all members' tasks must be finished one day prior to the PA => 2 days before the PA deadline.

✔ **Quality:** All documents written and pull requests must be reviewed by all other members of the team through meetings.

✔ **Tools:** All tasks are on the task board with an owner and due date and will also be announced through communication channels; code goes through pull requests or will be merged when all team members are present.

3.6 Use of AI Tools

✔ **AI Tools:**

- Claude, Antigravity, Codex: Coding
- Perplexity + Gemini: Finding data + Documents.

✔ **Reviewing:**

- Each usage of AI will be reviewed by at least one other AI model and one other team member.

- Each team member keeps their own AI usage log and when the team prepares for the PA, all AI usage logs will be gathered into one.

3.7 Decision Making

When making decisions, due to the fact that our team has 5 members, a majority vote is a suitable solution. Furthermore, if one team member is absent from a meeting or some other circumstances that might leave us at a tie, both sides will provide evidence and thought processes for their decisions and after that, the team lead will decide the outcome of the discussion.

3.8 Handling Conflicts and Missed Work

✔ **Step 1:** Talk directly and privately with the member and agree on a fix and a new date.

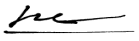
✔ **Step 2:** Make sure it doesn't affect the team's momentum by fixing the conflict or putting in extra hours to make up for missed work.

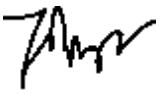
✔ **Step 3**: Discuss in a team meeting and record it in the weekly status report.

✔ **Step 4:** Inform the TA; the issue is taken into account in the individual contribution factor.

3.9 Signatures

By signing, each member agrees to follow this contract.

Student ID	Full name	Signature	Date
24127278	Hoàng Minh Huy		12/10/2026
24127048	Nguyễn Thế Huy		12/10/2026
24127450	Nguyễn Chí Minh		12/10/2026

24127329	Lê Nguyễn Quốc Bảo		12/10/2026
24127142	Đoàn Minh Tuấn		12/10/2026

4. AI Usage Log

No AI tools were used in this PA.

#	Member	Tool	Task / purpose	How the output was verified and changed